

ASSIGNMENT 4

Cryptanalysis of Shift Cipher using Brute Force, Dictionary Scoring and Chi-Square Analysis

Group-15

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1. Aim

To perform cryptanalysis of a Shift Cipher using brute-force search, dictionary scoring, and Chi-Square analysis, and to compare the keys predicted by the two cryptanalysis techniques.

2. Objectives

- Implement encryption and decryption for a Shift Cipher.
- Try all possible keys using brute-force cryptanalysis.
- Use an English dictionary to score possible plaintexts.
- Use Chi-Square analysis based on English letter frequencies.
- Compare the predicted keys with the actual encryption key.
- Analyze cases where an attack succeeds or fails.

3. Introduction

A Shift Cipher, also known as a Caesar Cipher, replaces each alphabetic character by shifting it a fixed number of positions in the alphabet. The encryption key is an integer from 0 to 25. Because there are only 26 possible keys, the cipher can be attacked by trying every key and evaluating the resulting plaintext.

4. Shift Cipher Implementation

The Shift Cipher was implemented in Python in the file `attacks/shift_cipher_attack/src/shift_cipher.py`. The `encrypt()` function shifts alphabetic characters by the given key, while `decrypt()` calls the same function with the negative key. Uppercase and lowercase letters are preserved, and spaces and punctuation remain unchanged.

5. Experimental Verification of Shift Cipher

The implementation was tested using the plaintext "hello world is my new doc." with key 3.

Parameter	Value
-----------	-------

Plaintext	hello world is my new doc.
Key	3
Encrypted text	khoor zruog lv pb qhz grf.
Decrypted text	hello world is my new doc.

The encrypted output and subsequent decryption matched the expected result, confirming that the Shift Cipher implementation works correctly.

6. Brute-Force Cryptanalysis

In brute-force cryptanalysis, all 26 possible keys are tested. For each key, the ciphertext is decrypted and the resulting plaintext is evaluated. Since the Shift Cipher has only 26 possible keys, exhaustive search is computationally simple.

The implemented dictionary-scoring approach loads words from `english_words.txt` and counts how many decrypted words match entries in the dictionary. The candidate with the highest score is selected as the predicted plaintext and key.

7. Dictionary Scoring Algorithm

1. Load the English dictionary into a set for fast word lookup.
2. Try every key from 0 to 25.
3. Decrypt the ciphertext using the current key.
4. Split the plaintext into words and compare them with the dictionary.
5. Increase the score for every matching dictionary word.
6. Rank all candidates by their dictionary score.
7. Select the candidate with the highest score.

8. Chi-Square Cryptanalysis

Chi-Square analysis uses the frequency of letters in English. For each possible key, the ciphertext is decrypted and the observed frequency of each letter is compared with the expected English frequency. A Chi-Square statistic is calculated for each candidate. The key producing the smallest Chi-Square value is selected because its letter distribution is closest to normal English.

9. Failure Analysis

An attack may fail when the ciphertext is too short to provide enough statistical information. Dictionary scoring may also fail when the plaintext contains words that are not present in the dictionary, spelling variations, names, or punctuation that is not handled by the scoring method. Chi-Square analysis can be unreliable for very short messages because the observed letter frequencies may differ substantially from average English frequencies.

Possible improvements include using a larger and more representative English dictionary, assigning different weights to common words, using n-gram language models, and testing the attacks on longer ciphertexts. Chi-Square performance can also be improved by using sufficiently long English text and a carefully selected frequency table.

10. Observations

- The Shift Cipher is weak because its key space contains only 26 possible keys.
- Brute-force search can generate every possible plaintext without knowing the key.
- Dictionary scoring can identify readable plaintext when enough recognizable English words are present.
- Chi-Square analysis uses statistical properties of English rather than complete-word matching.
- The reliability of both automated attacks depends on the length and characteristics of the ciphertext.
- A larger dictionary and better statistical model can improve cryptanalysis accuracy

11.

Screenshots and Experimental Evidence

```
dhillionice@dhillionice:~/crypto/lab4$ cd shift_cipher_attack/src
python3 main.py
=====
SHIFT CIPHER CRYPTANALYSIS
=====

Enter plaintext: this is a secret message
Enter encryption key (0-25): 3

Ciphertext:
wklv lv d vhfuhw phvvdjh

=====
DICTIONARY SCORING ATTACK
=====

Predicted Key: 3
Predicted Plaintext: this is a secret message
Dictionary Score: 5

=====
CHI-SQUARE ATTACK
=====

Predicted Key: 3
Predicted Plaintext: this is a secret message
Chi-Square Score: 21.6

=====
COMPARISON
=====

Actual Key: 3
Dictionary Correct? True
Chi-Square Correct? True
```

```
dhillonice@dhillonice:~/crypto/lab4/shift_cipher_attack/src$ python3 main.py
```

```
=====
SHIFT CIPHER CRYPTANALYSIS
=====
```

```
Enter plaintext: cryptography is important for computer security
Enter encryption key (0-25): 15
```

```
Ciphertext:
rgneidvgpewn xh xbedgipci udg rdbejitg htrjgxin
```

```
=====
DICTIONARY SCORING ATTACK
=====
```

```
Predicted Key: 15
Predicted Plaintext: cryptography is important for computer security
Dictionary Score: 4
```

```
=====
CHI-SQUARE ATTACK
=====
```

```
Predicted Key: 15
Predicted Plaintext: cryptography is important for computer security
Chi-Square Score: 39.17
```

```
=====
COMPARISON
=====
```

```
Actual Key: 15
Dictionary Correct? True
Chi-Square Correct? True
```

```
1  SHIFT CIPHER CRYPTANALYSIS RESULTS
2
3  Test Case 1
4  Plaintext: this is a secret message
5  Actual Key: 3
6  Dictionary Key: 3
7  Chi-Square Key: 3
8  Dictionary Correct: True
9  Chi-Square Correct: True
10
11
12  Test Case 2
13  Plaintext: hello world this is a test
14  Actual Key: 7
15  Dictionary Key: 7
16  Chi-Square Key: 7
17  Dictionary Correct: True
18  Chi-Square Correct: True
19
20
21  Test Case 3
22  Plaintext: cryptography is important for computer security
23  Actual Key: 15
24  Dictionary Key: 15
25  Chi-Square Key: 15
26  Dictionary Correct: True
27  Chi-Square Correct: True
28
29
30  Test Case 4
31  Plaintext: hello
32  Actual Key: 5
33  Dictionary Key: 5
34  Chi-Square Key: 16
35  Dictionary Correct: True
36  Chi-Square Correct: False
```

12. Conclusion

The assignment demonstrates how a Shift Cipher can be cryptanalyzed using exhaustive search and properties of the English language. The limited key space makes brute-force search practical, while dictionary scoring and Chi-Square analysis help select the most likely plaintext automatically. The experiment also shows that cryptanalysis accuracy depends on the amount and nature of available ciphertext. Overall, the Shift Cipher provides a clear example of why simple substitution methods with small key spaces are not secure.

13. Repository Structure

The assignment follows the required structure specified in the laboratory instructions:

```
CryptoLabX/
├── attacks/
│   ├── shift_cipher_attack/
│   │   ├── src/
│   │   │   ├── shift_cipher.py
│   │   │   ├── brute_force_dictionary.py
│   │   │   ├── chi_square_attack.py
│   │   │   └── main.py
│   │   ├── dictionary/
│   │   │   └── english_words.txt
│   │   ├── testcases/
│   │   ├── outputs/
│   │   ├── screenshots/
│   │   ├── reports/
│   │   │   └── Assignment_4_Report.pdf
│   └── README.md
```